

# Teaching Precommitted Agents: Model-Free Policy Evaluation and Control in Quasi-Hyperbolic Discounted MDPs

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**Abstract**—Time-inconsistent preferences arise in many real-world decision-making problems, where agents prefer smaller-sooner rewards over larger-later ones. Quasi-Hyperbolic (QH) discounting has emerged as a simple yet powerful model to capture such behavior. While extensively studied in behavioral economics, its implications in reinforcement learning (RL) remain underexplored. In this work, we investigate the RL problem under QH discounting with a focus on precommitted agents, who aim to maximize cumulative utility while being aware of their time-inconsistent preferences. We make two key contributions: (i) we formally characterize the structure of optimal policies under QH discounting and show, for the first time, that the optimal policy is a one-step non-stationary policy; (ii) we design model-free algorithms for both policy evaluation and policy optimization in this setting. Our results provide foundational insights for incorporating QH preferences in RL and open up new directions for time-inconsistent decision-making in sequential domains.

## I. INTRODUCTION

Reinforcement learning (RL) [11], [2] provides a powerful framework for sequential decision-making, where an agent interacts with an environment to maximize long-term cumulative rewards. A foundational assumption in most RL algorithms is that agents discount future rewards using an exponential discount factor, which implies time-consistent preferences: an agent's relative preference between two future rewards remains unchanged over time. However, this assumption does not hold in many real-world scenarios involving humans, animals, or even artificial agents facing bounded rationality or changing preferences [5].

Quasi-Hyperbolic (QH) discounting [10], [8], introduced in behavioral economics, offers a more realistic model of decision-making. Under QH discounting, agents apply a present-bias parameter  $\sigma \in [0, 1]$  in addition to the standard exponential discount factor  $\gamma \in [0, 1]$ . Immediate rewards are weighted more heavily than future rewards, capturing the observed preference for smaller-sooner outcomes over larger-later ones. Formally, the cumulative discounted return for a sequence of rewards  $(r_0, r_1, r_2, \dots)$  is given by:

$$G = r_0 + \sigma \sum_{t=1}^{\infty} \gamma^t r_t.$$

While QH discounting has been used to explain human behavior in intertemporal choice tasks, its incorporation into the RL framework poses unique challenges. Unlike exponential discounting, QH discounting leads to time-inconsistent

preferences: the optimal policy from the perspective of the present self may differ from what a future self would prefer. This gives rise to a spectrum of agent behaviors [9]:

- **Naive agents** fail to recognize their future time inconsistency and continually replan.
- **Sophisticated agents** anticipate their future selves' behavior and seek subgame perfect equilibria [7], [6].
- **Precommitted agents** understand their time inconsistency but fix their policies in advance to optimize cumulative QH-discounted return.

In this work, we focus on precommitted agents, who plan ahead and commit to a policy to maximize their full QH-discounted objective. This formulation allows us to treat the problem as a well-defined optimization problem and study the structure and computation of optimal policies under QH discounting.

*a) Contributions:* We make the following contributions:

- 1) We formally define the RL problem under QH discounting for precommitted agents and characterize the structure of optimal policies. To the best of our knowledge, we are the first to show that the optimal policy is a *one-step non-stationary policy*. That is, the optimal policy can be decomposed into a pair  $(\mu^*, \bar{\pi}^*)$ , where  $\mu^*$  is applied at the first step, and  $\bar{\pi}^*$  is a stationary policy used thereafter.
- 2) We provide model-free algorithms for evaluating the value function of a given policy under QH discounting. These algorithms use two-time-scale stochastic approximation methods.
- 3) We design a model-free algorithm to find the optimal policy under QH discounting. By separately estimating the Q-functions corresponding to exponential and QH discounting, we derive a simple off-policy learning algorithm that converges to the optimal one-step non-stationary policy.

Our work bridges the gap between behavioral models of discounting and practical RL algorithms, enabling the development of agents that better align with human-like preferences in long-term decision-making tasks. The results presented here lay the groundwork for future studies in time-inconsistent RL and provide new tools for designing agents that account for dynamic preferences.

*b) Organization:* The remainder of the paper is structured as follows. In Section II, we introduce the MDP model with QH discounting and discuss the concept of nonstationary policies and their one-step equivalent represen-

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tation. Section III presents the policy evaluation algorithm for a given one-step non-stationary policy. In Section IV, we analyze the structure of the optimal policy and propose a model-free algorithm for its computation. In Section V, we demonstrate the effectiveness of the proposed algorithm by applying it to the standard inventory control problem to compute the optimal policy.

## II. SETUP AND PRELIMINARIES

### A. Setup

Let  $\Delta(\mathcal{U})$  denote the set of probability distributions over a set  $\mathcal{U}$ . Our setup consists of a Markov Decision Process (MDP)  $M \equiv (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \sigma, \gamma)$ , where:  $\mathcal{S}$  is a finite state space,  $\mathcal{A}$  is a finite action space,  $\mathcal{P} : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$  is the transition probability kernel,  $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$  is the reward function,  $\sigma, \gamma \in [0, 1)$  are the parameters of quasi-hyperbolic (QH) discounting.

We denote the quasi-hyperbolic value functions corresponding to a policy  $\nu$  by  $V_\nu^{\sigma, \gamma}$  and  $Q_\nu^{\sigma, \gamma}$ , and the standard exponentially discounted value functions for the same policy by  $V_\nu^\gamma$  and  $Q_\nu^\gamma$ , respectively. Under QH discounting, the value function associated with a policy  $\nu = (\nu_0, \nu_1, \dots)$  can be rewritten as

$$V_\nu^{\sigma, \gamma}(s) = \mathbb{E}_{\substack{a_0 \sim \nu_0(\cdot|s) \\ s' \sim P(\cdot|s, a_0)}} \left[ r(s, a_0) + \sigma \gamma V_{(\nu_1, \nu_2, \dots)}^\gamma(s') \right]$$

### B. One-Step Non-Stationary Representation

A well-known fact in exponential discounting is that the value function of any non-stationary policy is equivalent to the value function of a corresponding stationary policy. In a similar spirit, we establish a new result for quasi-hyperbolic discounting: any non-stationary policy can be equivalently represented as a *one-step non-stationary policy*, where a distinct policy  $\mu$  is followed in the first step, after which a stationary policy  $\pi$  is followed indefinitely.

**Theorem 1.** *For a finite-state, finite-action MDP, the value function under quasi-hyperbolic discounting of any non-stationary policy  $\nu = (\nu_0, \nu_1, \dots)$  is equal to the value function corresponding to a one-step non-stationary policy  $(\mu, \bar{\pi})$  for some policies  $\mu$  and  $\bar{\pi}$ .*

*Proof.* From the definition of QH-discounted value functions,

$$V_\nu^{\sigma, \gamma}(s) = \mathbb{E} \left[ \sum_{t=0}^{\infty} d(t) r(s_t, a_t) \mid s_0 = s \right],$$

where the QH discount factor  $d(t)$  is given by

$$d(t) = \begin{cases} 1, & t = 0, \\ \sigma \gamma^t, & t \geq 1. \end{cases}$$

Expanding the expectation, we obtain:

$$V_\nu^{\sigma, \gamma}(s) = \mathbb{E} \left[ r(s_0, a_0) + \sum_{t=1}^{\infty} \sigma \gamma^t r(s_t, a_t) \mid s_0 = s \right].$$

Rewriting the expectation by conditioning on the first action  $a_0 \sim \nu_0(\cdot|s)$ , we get:

$$V_\nu^{\sigma, \gamma}(s) = \mathbb{E} \left[ r(s, a_0) + \mathbb{E} \left[ \sum_{t=1}^{\infty} \sigma \gamma^t r(s_t, a_t) \mid s_0, a_0 \right] \right].$$

Further conditioning on the next state  $s' \sim P(\cdot|s, a_0)$ , we obtain:

$$V_\nu^{\sigma, \gamma}(s) = \mathbb{E} \left[ r(s, a_0) + \mathbb{E} \left[ \sum_{t=1}^{\infty} \sigma \gamma^t r(s_t, a_t) \mid s_1 = s' \right] \right].$$

Using the fact that the remaining discounted sum follows an exponentially discounted value function,

$$\mathbb{E} \left[ \sum_{t=1}^{\infty} \sigma \gamma^t r(s_t, a_t) \mid s_1 = s' \right] = \sigma \gamma V_{(\nu_1, \nu_2, \dots)}^\gamma(s'),$$

we conclude:

$$V_\nu^{\sigma, \gamma}(s) = \mathbb{E}_{\substack{a_0 \sim \nu_0(\cdot|s) \\ s' \sim P(\cdot|s, a_0)}} \left[ r(s, a_0) + \sigma \gamma V_{(\nu_1, \nu_2, \dots)}^\gamma(s') \right].$$

It is well established that under standard exponential discounting, there exists a stationary policy whose value function matches that of any given non-stationary policy [1]. Let  $\bar{\pi}$  be a stationary policy such that:

$$V_{(\nu_1, \nu_2, \dots)}^\gamma(s) = V_{\bar{\pi}}^\gamma(s), \quad \forall s \in \mathcal{S}.$$

Substituting this into the previous equation, we obtain:

$$V_\nu^{\sigma, \gamma}(s) = \sum_a \nu_0(a|s) \left[ r(s, a) + \sigma \gamma \sum_{s'} P(s'|s, a) V_{\bar{\pi}}^\gamma(s') \right].$$

Defining  $\mu = \nu_0$  as the first-step policy, we conclude that:

$$V_\nu^{\sigma, \gamma}(s) = V_{(\mu, \bar{\pi})}^{\sigma, \gamma}(s),$$

thus proving the theorem.  $\square$

## III. POLICY EVALUATION

In this section, we present a model-free algorithm for evaluating the value function of a one-step nonstationary policy  $(\mu, \bar{\pi})$  under quasi-hyperbolic (QH) discounting. As established in the previous section, it suffices to work with such policies instead of general nonstationary policies. However, unlike in exponential discounting, the value function  $V_{\mu, \bar{\pi}}^{\sigma, \gamma}$  does not admit a simple recursive formulation. Instead, we express  $V_{\mu, \bar{\pi}}^{\sigma, \gamma}$  in terms of  $V_{\bar{\pi}}^{\sigma, \gamma}$ , which itself satisfies a recursive equation as follows:

$$\begin{aligned} V_{\mu, \bar{\pi}}^{\sigma, \gamma}(s) &= \sum_a \mu(a|s) \left[ r(s, a) + \sum_{s'} P(s'|s, a) \left[ - (1 - \sigma) \gamma \right. \right. \\ &\quad \left. \left. \sum_{a'} \pi(a'|s') r(s', a') + \gamma V_{\bar{\pi}}(s') \right] \right], \\ V_{\bar{\pi}}^{\sigma, \gamma}(s) &= \sum_a \pi(a|s) \left[ r(s, a) + \sum_{s'} P(s'|s, a) \left[ - (1 - \sigma) \gamma \right. \right. \\ &\quad \left. \left. \sum_{a'} \pi(a'|s') r(s', a') + \gamma V_{\bar{\pi}}(s') \right] \right]. \end{aligned} \tag{1}$$

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**Algorithm 1** Model-Free Policy Evaluation for QH Discounting

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1: Input: Stepsizes  $(\alpha_n), (\beta_n)$ ; discount factors  $\sigma, \gamma$ ; sampling policy  $\nu$ ; evaluation policies  $\mu, \pi$ 
2: Initialize:  $V_0, W_0 \in \mathbb{R}^{|\mathcal{S}|}$ 
3: for  $n = 0, 1, 2, \dots$  do
4:   for  $s \in \mathcal{S}$  do
5:     Sample  $a \sim \nu(\cdot|s)$ 
6:     Observe reward  $r(s, a)$  and next state  $s' \sim P(\cdot|s, a)$ 
7:     Sample  $a' \sim \pi(\cdot|s')$ 
8:     Observe reward  $r(s', a')$ 
9:     Compute TD target:
        
$$r_n^{\text{target}}(s) \leftarrow r(s, a) - (1 - \sigma)\gamma r(s', a') + \gamma W_n(s')$$

10:    Update  $W$ :
        
$$W_{n+1}(s) \leftarrow W_n(s) + \alpha_n \left[ \frac{\pi(a|s)}{\nu(a|s)} r_n^{\text{target}}(s) - W_n(s) \right]$$

11:    Update  $V$ :
        
$$V_{n+1}(s) \leftarrow V_n(s) + \beta_n \left[ \frac{\mu(a|s)}{\nu(a|s)} r_n^{\text{target}}(s) - V_n(s) \right]$$

12:   end for
13: end for

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Since the evaluation of both  $V_{\mu, \pi}^{\sigma, \gamma}$  and  $V_{\pi, \pi}^{\sigma, \gamma}$  requires sampling actions according to the policies  $\mu$  and  $\pi$  respectively, we introduce an alternative approach using a sampling policy  $\nu$ . Instead of directly sampling from  $\mu$  or  $\pi$ , we sample actions from  $\nu$  and apply importance sampling corrections to adjust the updates accordingly.

To approximate the value functions, we maintain two iterates: (i)  $W_n(s)$ , which approximates  $V_{\pi, \pi}^{\sigma, \gamma}(s)$ , and (ii)  $V_n(s)$ , which approximates  $V_{\mu, \pi}^{\sigma, \gamma}(s)$ . See Algorithm 1 for more details.

The step sizes  $\alpha_n$  and  $\beta_n$  follow the standard Robbins-Monro conditions. These updates ensure that both  $W_n(s)$  and  $V_n(s)$  converge to their respective value functions (see Theorem 2) while efficiently leveraging off-policy sampling. We employ a sampling policy  $\nu$  that satisfies  $\nu(a|s) > 0$  for all state-action pairs  $(s, a)$ . If either  $\mu$  or  $\pi$  also satisfy this condition, they can be used as the sampling policy instead. By incorporating the importance sampling correction terms  $\frac{\pi(a|s)}{\nu(a|s)}$  and  $\frac{\mu(a|s)}{\nu(a|s)}$ , we obtain unbiased estimates of expectations with respect to the target policies  $\pi$  and  $\mu$ , despite sampling actions according to  $\nu$ .

We now provide results related to policy evaluation in QH discounting for one-step non stationary policies. We start off with a Bellman type operator for stationary policy in QH discounting. Then provide the theorem which talks about the convergence of the proposed algorithm.

**Lemma 1.** Define the operator  $T_{\bar{\pi}} : \mathbb{R}^{|\mathcal{S}|} \rightarrow \mathbb{R}^{|\mathcal{S}|}$  as follows:

$$(T_{\bar{\pi}}V)(s) = \sum_a \pi(a|s) \left[ r(s, a) + \sum_{s'} P(s'|s, a) \left[ - (1 - \sigma)\gamma \sum_{a'} \pi(a'|s') r(s', a') + \gamma V(s') \right] \right]. \quad (2)$$

Then,  $T_{\bar{\pi}}$  is a contraction mapping under the max norm, and the unique fixed point of  $T_{\bar{\pi}}$  is the value function  $V_{\bar{\pi}}^{\sigma, \gamma}$ .

*Proof.* We first show that  $T_{\bar{\pi}}$  is a contraction with respect to the supremum norm  $\|\cdot\|_{\infty}$ . Let  $V_1, V_2 \in \mathbb{R}^{|\mathcal{S}|}$  be any two vectors. Consider the difference:

$$|(T_{\bar{\pi}}V_1)(s) - (T_{\bar{\pi}}V_2)(s)|. \quad (3)$$

Expanding the operator definition, we obtain:

$$\begin{aligned} & |(T_{\bar{\pi}}V_1)(s) - (T_{\bar{\pi}}V_2)(s)| \\ &= \left| \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) \gamma (V_1(s') - V_2(s')) \right| \\ &\leq \gamma \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) |V_1(s') - V_2(s')| \\ &\leq \gamma \|V_1 - V_2\|_{\infty}. \end{aligned}$$

Taking the maximum over all states  $s$ , we obtain:

$$\|T_{\bar{\pi}}V_1 - T_{\bar{\pi}}V_2\|_{\infty} \leq \gamma \|V_1 - V_2\|_{\infty}. \quad (4)$$

Since  $\gamma \in [0, 1)$ , this implies that  $T_{\bar{\pi}}$  is a contraction mapping with contraction factor  $\gamma$ .

By Banach's fixed-point theorem,  $T_{\bar{\pi}}$  has a unique fixed point  $V^*$ , i.e.,  $T_{\bar{\pi}}V^* = V^*$ . Observing the original Bellman equation, we conclude that the unique fixed point is precisely the value function  $V_{\bar{\pi}}^{\sigma, \gamma}$ .

Thus, the value function  $V_{\bar{\pi}}^{\sigma, \gamma}$  is the unique solution to  $V = T_{\bar{\pi}}V$ , completing the proof.  $\square$

We now present the main result of this section, establishing the convergence of the proposed policy evaluation algorithm under standard assumptions.

**Theorem 2** (Convergence of Policy Evaluation). *Let the sequences  $(W_n, V_n)$  be updated according to Algorithm 1, and suppose the step sizes satisfy the Robbins-Monro conditions:*

$$\sum_n \alpha_n = \infty, \quad \sum_n \alpha_n^2 < \infty, \quad \sum_n \beta_n = \infty, \quad \sum_n \beta_n^2 < \infty.$$

*Further, assume a two-time-scale setting where*

$$\frac{\beta_n}{\alpha_n} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

*Then, the iterates  $(W_n, V_n)$  converge almost surely to  $(V_{\bar{\pi}}^{\sigma, \gamma}, V_{\mu, \pi}^{\sigma, \gamma})$ .*

*Proof Sketch.* The proof follows the standard two-time-scale stochastic approximation framework as developed in [3]. The key idea is to analyze the evolution of the fast and slow iterates separately, by treating one as quasi-static while analyzing the other.

a) *Step 1: Fast Time-Scale Convergence of  $W_n$* : On the fast time scale,  $W_n$  evolves according to

$$W_{n+1}(s) = W_n(s) + \alpha_n [T_{\bar{\pi}} W_n(s) - W_n(s) + \text{noise}].$$

The noise sequence forms a martingale difference and satisfies the usual boundedness and moment assumptions. Thus, by standard results on stochastic approximation with contraction operators (see [3]), we conclude that  $W_n \rightarrow V_{\bar{\pi}}^{\sigma, \gamma}$  almost surely.

b) *Step 2: Convergence of the Slow Time-Scale Iterates  $V_n$* : On the slower time scale (governed by  $\beta_n$ ), we treat  $W_n \approx V_{\bar{\pi}}^{\sigma, \gamma}$  as approximately converged and quasi-static. Then, the expected update defines a linear dynamical system whose unique fixed point is  $V_{\mu^*, \bar{\pi}}^{\sigma, \gamma}(s)$ . Therefore,  $V_n \rightarrow V_{\mu^*, \bar{\pi}}^{\sigma, \gamma}$  almost surely.  $\square$

#### IV. OPTIMAL CONTROL

In this section, we define the optimal policy under quasi-hyperbolic (QH) discounting and provide a model-free algorithm to compute it.

**Definition 1.** *The optimal policy in QH discounting is a policy  $(\mu^*, \bar{\pi}^*)$  that maximizes the value function for all states, i.e.,*

$$V_{\mu^*, \bar{\pi}^*}^{\sigma, \gamma}(s) = \max_{\mu, \bar{\pi}} V_{\mu, \bar{\pi}}^{\sigma, \gamma}(s), \quad \forall s. \quad (5)$$

To derive the structure of the optimal policy, we consider

$$\begin{aligned} V_{\mu^*, \bar{\pi}^*}^{\sigma, \gamma}(s) &= \max_{\mu, \bar{\pi}} V_{\mu, \bar{\pi}}^{\sigma, \gamma}(s) \\ &= \max_{\mu, \bar{\pi}} \sum_a \mu(a|s) \left[ r(s, a) + \sigma \gamma \sum_{s'} P(s'|s, a) V_{\bar{\pi}}^{\gamma}(s') \right] \\ &= \max_{\mu} \sum_a \mu(a|s) \left[ r(s, a) + \sigma \gamma \sum_{s'} P(s'|s, a) \max_{\bar{\pi}} V_{\bar{\pi}}^{\gamma}(s') \right] \end{aligned}$$

Since  $\max_{\bar{\pi}} V_{\bar{\pi}}^{\gamma}(s)$  corresponds to the optimal value function under exponential discounting, we denote the corresponding optimal policy as  $\pi_{\gamma}^*$ . Consequently, the optimal QH-discounted policy is given by

$$\bar{\pi}^* = \pi_{\gamma}^*,$$

and the optimal  $\mu^*$  satisfies

$$\begin{aligned} \mu^*(s) &= \arg \max_{\mu} \sum_a \mu(a|s) \left[ r(s, a) \right. \\ &\quad \left. + \sigma \gamma \sum_{s'} P(s'|s, a) V_{\pi_{\gamma}^*}^{\gamma}(s') \right]. \end{aligned} \quad (6)$$

**Theorem 3.** *There exists an optimal policy  $(\mu^*, \bar{\pi}^*)$  that is deterministic. That is,*

$$V_{\mu^*, \bar{\pi}^*}^{\sigma, \gamma}(s) = \max_{\mu, \bar{\pi}} V_{\mu, \bar{\pi}}^{\sigma, \gamma}(s), \quad \forall s. \quad (7)$$

Hence, it is sufficient to restrict our search to deterministic policies.

*Proof.* We first show that  $\bar{\pi}^*$  can be chosen to be deterministic. From standard reinforcement learning theory, the optimal policy under exponential discounting,  $\pi_{\gamma}^*$ , is known to be deterministic. Since the optimal  $\bar{\pi}^*$  in quasi-hyperbolic discounting is exactly  $\pi_{\gamma}^*$ , it follows that  $\bar{\pi}^*$  is deterministic.

Next, we show that  $\mu^*$  is also deterministic. The optimization over  $\mu$  is given by

$$\begin{aligned} \mu^*(s) &= \arg \max_{\mu} \sum_a \mu(a|s) \left[ r(s, a) \right. \\ &\quad \left. + \sigma \gamma \sum_{s'} P(s'|s, a) V_{\bar{\pi}^*}^{\gamma}(s') \right]. \end{aligned}$$

Since  $\bar{\pi}^*$  is deterministic, the value function  $V_{\bar{\pi}^*}^{\gamma}(s)$  is well-defined. The term inside the expectation is a linear function of  $\mu$ , and since  $\mu$  is a probability distribution over actions, the maximization corresponds to a standard linear program over a simplex. It is well known that the optimal solution to such a problem occurs at an extreme point, meaning that  $\mu^*(s)$  places all probability mass on a single action for each  $s$ . Thus,  $\mu^*$  is also deterministic.

This proves the existence of a deterministic optimal policy  $(\mu^*, \bar{\pi}^*)$ .  $\square$

##### A. Value Iteration Algorithm

In the quasi-hyperbolic discounted setting, the optimal policy can be derived through a simple modification of the standard value iteration procedure. Since the optimal long-term component of the policy corresponds to the exponential discounting case, we first compute the optimal exponential value function  $V_{\pi_{\gamma}^*}^{\gamma}$  using standard value iteration. Once  $V_{\pi_{\gamma}^*}^{\gamma}$  is obtained, the precommitted optimal policy  $\mu^*$  for the initial step can be computed using (6).

Thus, the value iteration procedure for QH optimal control consists of:

- 1) Performing standard value iteration to compute  $V_{\pi_{\gamma}^*}^{\gamma}$ .
- 2) Using (6) to compute the optimal one-step policy  $\mu^*$ .

This two-step procedure efficiently computes the optimal policy for a precommitted agent under QH discounting.

When the model is known, finding the optimal policy becomes straightforward once the value function  $V_{\pi_{\gamma}^*}^{\gamma}$  is computed. However, in a model-free setting, it is more practical to work with Q-value functions, which allow us to estimate the optimal policy without explicit knowledge of the system dynamics. The model-free approach based on Q-values is discussed in the next subsection.

##### B. Model-Free Algorithm

In this subsection, we present a model-free algorithm to compute the optimal policy in quasi-hyperbolic (QH) discounting. The Q-function for a given policy  $(\mu, \bar{\pi})$  satisfies the following relationship:

$$Q_{\mu, \bar{\pi}}^{\sigma, \gamma}(s, a) = (1 - \sigma)r(s, a) + \sigma Q_{\mu, \bar{\pi}}^{\gamma}(s, a). \quad (8)$$

Since the long-term component of the Q-function,  $Q_{\mu, \bar{\pi}}^{\gamma}(s, a)$ , does not depend on the one-step non-stationary policy  $\mu$ , we can write

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**Algorithm 2** QH Q-learning (Model-Free)

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**Input:** Stepsizes  $(\alpha_n), (\beta_n)$ , discount factor  $\gamma$ , QH parameter  $\sigma$

**Initialize:**  $W_0, Q_0 \in \mathbb{R}^{|S \times A|}$

**for**  $n = 0, 1, 2, \dots$  **do**

**for**  $(s, a) \in \mathcal{S} \times \mathcal{A}$  **do**

        Sample  $s' \sim P(\cdot | s, a)$

        Observe reward  $r(s, a)$

$$W_{n+1}(s, a) \leftarrow W_n(s, a) + \alpha_n [r(s, a) + \gamma \max_{a'} W_n(s', a') - W_n(s, a)]$$

$$Q_{n+1}(s, a) \leftarrow Q_n(s, a) + \beta_n [(1 - \sigma)r(s, a) + \sigma W_n(s, a) - Q_n(s, a)]$$

**end for**

**end for**

**Output:** QH value function  $Q_n$ , exponential value function  $W_n$  and optimal policy

$$\mu^*(s) = \arg \max_a Q(s, a),$$

$$\pi^*(s) = \arg \max_a W(s, a).$$


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$$Q_{\mu, \pi}^\gamma(s, a) = Q_\pi^\gamma(s, a).$$

Now, we express the optimal Q-function as

$$\begin{aligned} Q_{\mu^*, \pi^*}^{\sigma, \gamma}(s, a) &= \max_{\mu, \pi} [(1 - \sigma)r(s, a) + \sigma Q_{\mu, \pi}^\gamma(s, a)] \\ &= \max_{\pi} [(1 - \sigma)r(s, a) + \sigma Q_\pi^\gamma(s, a)] \\ &= (1 - \sigma)r(s, a) + \sigma \max_{\pi} Q_\pi^\gamma(s, a). \end{aligned}$$

Since it is well known that the optimal Q-function under exponential discounting satisfies

$$\max_{\pi} Q_\pi^\gamma(s, a) = Q_{\pi^*}^\gamma(s, a),$$

we conclude that the optimal policies in QH discounting are given by

$$\bar{\pi}^* = \pi_\gamma^*,$$

and

$$\begin{aligned} \mu^*(s) &= \arg \max_a Q_{\mu^*, \bar{\pi}^*}^{\sigma, \gamma}(s, a) \\ &= \arg \max_a [(1 - \sigma)r(s, a) + \sigma Q_{\bar{\pi}^*}^\gamma(s, a)]. \end{aligned}$$

*a) Algorithm:* To estimate the Q-value functions under both exponential and QH discounting, we maintain two iterates,  $W_n$  and  $Q_n$ , where: (i)  $W_n$  tracks the Q-function under exponential discounting. (ii)  $Q_n$  tracks the Q-function under QH discounting. See Algorithm 2 for more details.

This model-free approach effectively learns the optimal policies in QH discounting using off-policy learning. We now

provide result related to Q-learning and finding a one-step non stationary optimal policy in QH discounting.

**Theorem 4** (Convergence of QH Q-learning Algorithm). *Let  $Q_n$  and  $W_n$  be the sequences updated according to the QH Q-learning algorithm (Algorithm 2). Assume that the step sizes  $\alpha_n$  and  $\beta_n$  satisfy the Robbins-Monro conditions:*

$$\sum_n \alpha_n = \infty, \quad \sum_n \alpha_n^2 < \infty, \quad \sum_n \beta_n = \infty, \quad \sum_n \beta_n^2 < \infty,$$

*and the two-time-scale assumption:*

$$\frac{\beta_n}{\alpha_n} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

*Then, the iterates  $(Q_n, W_n)$  converge to the true QH Q-value function  $Q_{\mu^*, \pi^*}^{\sigma, \gamma}$  and the exponential value function  $Q_{\pi^*}^\gamma$ , respectively. Moreover, the optimal policy  $\mu^*$  and  $\pi^*$  are given by:*

$$\mu^*(s) = \arg \max_a Q_{\mu^*, \pi^*}^{\sigma, \gamma}(s, a), \quad \pi^*(s) = \arg \max_a Q_{\pi^*}^\gamma(s, a).$$

*Proof Sketch.* The proof follows the standard two-time-scale stochastic approximation framework as developed in [3]. The key idea is to analyze the evolution of the fast and slow iterates separately, by treating one as quasi-static while analyzing the other.

*b) Step 1: Convergence of  $W_n$ :* The update rule for  $W_n$  is exactly the Q-learning update rule in the exponential discounting case. The update rule can be recognized as a stochastic approximation update for a fixed-point iteration, which is known to converge to  $Q_{\pi^*}^\gamma$  almost surely [4].

*c) Step 2: Convergence of  $Q_n$ :* On the slower time scale (governed by  $\beta_n$ ), we treat  $W_n \approx Q_{\pi^*}^\gamma$  as approximately converged and quasi-static. Then, the expected update defines a linear dynamical system of the QH Q-value function for a precommitted agent. Therefore,  $Q_n$  iterates will converge to  $Q_{\mu^*, \pi^*}^{\sigma, \gamma}$ , almost surely.

*d) Step 3: Convergence of the Policies:* Once the Q-values  $Q_n$  and the exponential value functions  $W_n$  converge, the optimal policies  $\mu^*$  and  $\pi^*$  are given by:

$$\mu^*(s) = \arg \max_a Q_{\mu^*, \pi^*}^{\sigma, \gamma}(s, a), \quad \pi^*(s) = \arg \max_a W_{\pi^*}^\gamma(s, a).$$

Thus, we have shown that the sequences  $(Q_n, W_n)$  converge to the true value functions, and the optimal policies are determined accordingly.  $\square$

## V. EXPERIMENTS

### A. Inventory Control problem setup

In this subsection, we describe the inventory control problem which is used to run our simulations. We model the inventory control problem as an MDP, where the objective is to minimize the expected long-term cost.

The inventory has a maximum capacity of  $M$ , and the state of the system at any given time represents the number of items available in stock on that day. Thus, the state space is given by  $\mathcal{S} = \{0, 1, \dots, M\}$ . The action space is  $\mathcal{A} = \{0, 1, \dots, M\}$ , where an action  $a \in \mathcal{A}$  denotes the number of items procured on that day. The immediate cost consists

of three components: (i) the procurement cost incurred when purchasing items, (ii) the holding cost associated with storing unsold inventory, and (iii) the revenue generated from selling items. Let  $c$  denote the unit cost of purchasing an item,  $h$  represent the holding cost per unit of unsold inventory, and  $p$  denote the unit selling price. Let  $s_t, a_t$ , and  $d_t$  denote the state, action, and demand on day  $t$ . We define  $\hat{s}_t = \min(s_t + a_t, M)$ , the total number of items available on day  $t$  after procurement. Then, the next state is given by  $s_{t+1} = \max(\hat{s}_t - d_t, 0)$ , and the immediate cost incurred on day  $t$  is given by  $C(s_t, a_t, d_t) = c \times a_t + h \times \max(\hat{s}_t - d_t, 0) - p \times \min(\hat{s}_t, d_t)$ .

In this work, we consider an inventory system with parameters  $M = 2, c = 5, h = 2$  and  $p = 9$ . The daily demand is a random variable with values 0, 1, 2 with probabilities 0.2, 0.3 and 0.5. We consider the discount factors to be  $\sigma = 0.3$  and  $\gamma = 0.9$ .

### B. Optimal Policy

For the considered setup, we first computed the optimal exponential value function using the standard value iteration algorithm. The resulting optimal policy  $\pi^*$  was obtained by selecting greedy actions with respect to this exponential value function. Subsequently, the optimal pre-committed policy  $\mu^*$  was derived using Equation (6). This yielded the policy pair:  $\mu^*(0) = 1, \mu^*(1) = 0, \mu^*(2) = 0$  and  $\pi^*(0) = 2, \pi^*(1) = 1, \pi^*(2) = 0$ .

The same policy pair was recovered using our proposed QH Q-learning algorithm (Algorithm 2), validating the correctness of the model-free approach. Table I presents the learned QH Q-value function  $Q_{\mu^*, \pi^*}^{\sigma, \gamma}$  corresponding to the optimal policy.

TABLE I  
QH Q-VALUE FUNCTION FOR POLICY  $(\mu^*, \pi^*)$ .

| $s \backslash a$ | 0            | 1            | 2     |
|------------------|--------------|--------------|-------|
| 0                | 9.31         | <b>11.38</b> | 10.55 |
| 1                | <b>16.38</b> | 15.55        | 10.55 |
| 2                | <b>20.55</b> | 15.55        | 15.55 |

### C. Policy Evaluation

In this subsection, we evaluate the proposed policy evaluation algorithm (Algorithm 1) on the inventory control setup for three different policies, denoted as  $\rho = (\mu, \bar{\pi})$ , where  $\mu$  governs the action selection at time step  $t = 1$  and  $\bar{\pi}$  governs the policy for all subsequent time steps  $t > 1$ .

We consider the following three policy pairs for evaluation:

1.  $(\mu^*, \bar{\pi}^*)$  with an equiprobable sampling policy  $\nu$ .
2.  $(\mu^*, \bar{\pi}_u)$ , where  $\bar{\pi}_u$  is an equiprobable policy, using  $\bar{\pi}_u$  itself as the sampling policy  $\nu$ .
3.  $(\mu_u, \bar{\pi}^*)$ , where  $\mu_u$  is an equiprobable policy, using  $\mu_u$  as the sampling policy  $\nu$ .

Algorithm 1 is executed for each of these cases, and the results are analyzed by plotting  $\|V_k - V_{\rho}^{\sigma, \gamma}\|_2$  over iterations  $k$ . Figure 1 presents the results on a logarithmic scale along the x-axis. As seen in the figure,  $V_k$  converges

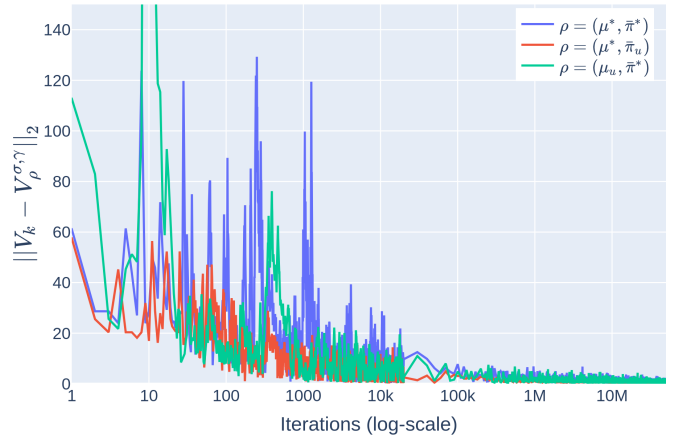


Fig. 1. Convergence of Policy Evaluation in Inventory Control

to  $V_{\rho}^{\sigma, \gamma}$  for each policy configuration, empirically validating our theoretical claims.

## VI. CONCLUSION

This paper explores quasi-hyperbolic (QH) discounting in Reinforcement Learning, analyzing its impact on policy evaluation and optimal control. We establish that any nonstationary policy can be equivalently represented as a one-step nonstationary policy, simplifying analysis and computation. We propose a model-free policy evaluation algorithm and a Q-learning algorithm for computing optimal policies in QH discounted MDPs, showing that the exponential value function updates align with standard Q-learning. Our results demonstrate that optimal policies integrate both exponential and QH discounting principles, providing practical tools for decision-making in model-free settings. Future work could extend these methods to more complex scenarios, such as POMDPs or multi-agent systems, where precommitment behavior plays a crucial role.

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